



CREATE AN INTERNAL NETWORK OF VIRTUAL MACHINES FOR PENETRATION TESTING

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Version: 1.0.0

Creation Date: 2026-06

Last Updated: 2026-06

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1 MAIN INFORMATION

1.1 Warning

The information and tools presented are intended for educational purposes only, to raise awareness and provide instruction. I would like to remind you that any use of the knowledge acquired must comply with the laws in force. The use of this knowledge for illegal or malicious activities is strictly forbidden and may result in legal consequences. I would like to make it clear that I cannot be held responsible for any illegal acts or behavior that may result from the use of this knowledge.

1.2 Course aim

Create an internal network of virtual machines for penetration testing.

1.3 Course requirements

- A type 2 hypervisor called VirtualBox.
- Basic knowledge of networks.
- Basic knowledge of linux operating system.

2 VIRTUAL MACHINES

2.1 Attacker's virtual machine

For the attacker's virtual machine, I'm going to use **Debian**. If you start out in the world of cyber security and have some knowledge of the Linux operating system, you can use **Kali Linux**, which is a Debian-based Linux distribution specially designed for penetration testing and security audits.

2.2 Target's virtual machine

For the target's virtual machine, I've chosen a virtual machine called **Kioptrix: Level 1**, which you can find on the **VulnHub** website.

3 CREATING AN INTERNAL NETWORK ON VIRTUALBOX

To create an internal network in virtualbox, you need to follow these steps :

- Go to your virtual machine's settings.
- Go to the **Network** menu.
- In the **Attached to** drop-down menu, select: **Internal Network**.
- In **Name** you must define a name for your network. In my case, I named it : **Isolated_Network**.
- Press **OK** to confirm.

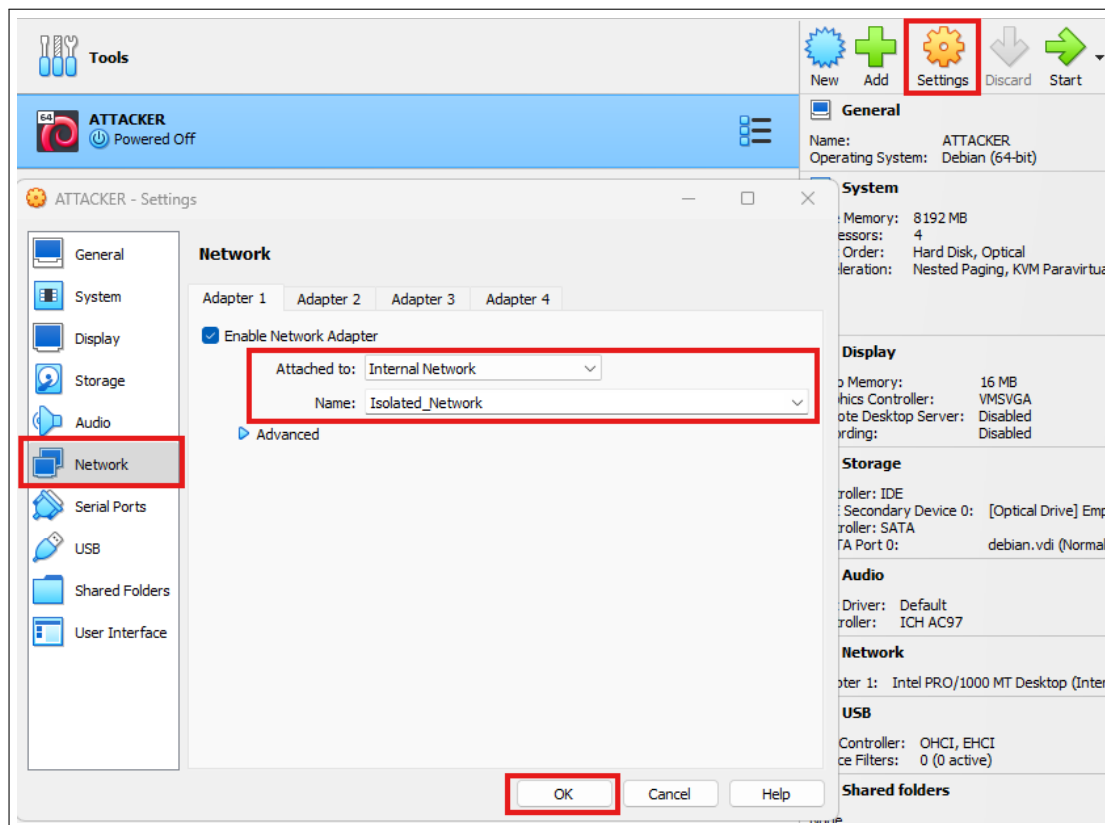


Figure 1: Virtual machine network settings

3.1 Network configuration of the attacker's virtual machine

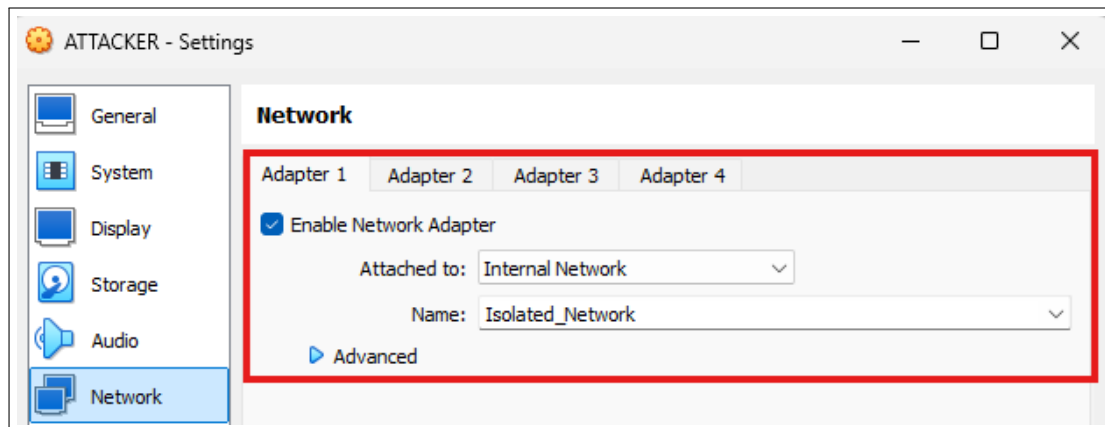


Figure 2: Attacker's virtual machine network settings

3.2 Network configuration of the target's virtual machine

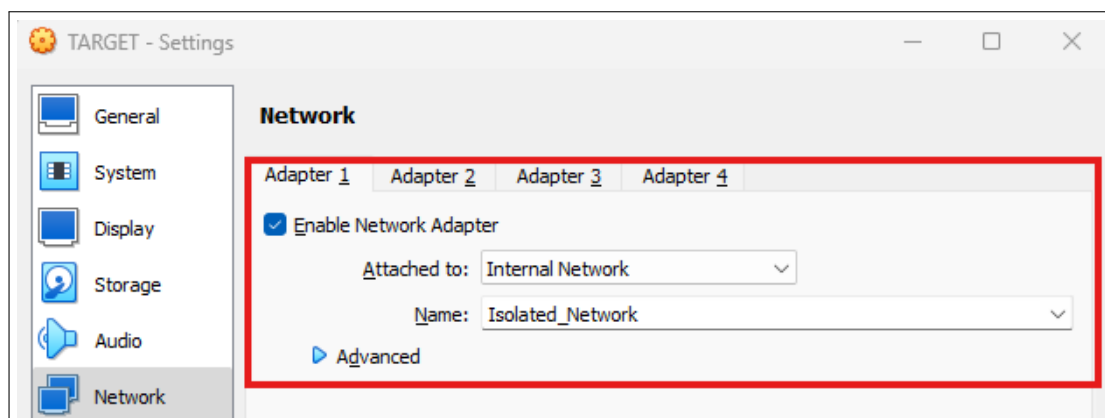


Figure 3: Target's virtual machine network settings

4 SETTING UP A DHCP SERVER

DHCP (Dynamic Host Configuration Protocol) is a network protocol that dynamically assigns a network configuration to the various devices (computer / phone / ...) that connect to the network.

4.1 DHCP server configuration

To create a DHCP server, we're going to use **VBoxManage** a command-line program that lets you manage various things in VirtualBox. This program is usually located in the folder where VirtualBox is installed. Here's the command we're going to use:

```
.\VBoxManage.exe dhcpserver add --network="Isolated_Network" --server-ip=10.0.0.2
--lower-ip=10.0.0.10 --upper-ip=10.0.0.20 --netmask=255.255.255.0 --enable
```

```
c:\Program Files\Oracle\VirtualBox>.\VBoxManage.exe dhcpserver add --network="Isolated Network"
--server-ip=10.0.0.2 --lower-ip=10.0.0.10 --upper-ip=10.0.0.20 --netmask=255.255.255.0 --enable
```

Figure 4: Command execution

TERM	DEFINITION
.\VboxManage.exe	A command-line program that lets you manage various things in virtualbox.
dhcpserver add	Add a DHCP server.
--network=""	The name of the network to which the DHCP server will be associated.
--server-ip=	The IP address of the DHCP server.
--lower-ip=	The lower IP address of the IP address range that the DHCP server can assign.
--upper-ip=	The upper IP address of the IP address range that the DHCP server can assign.
--netmask=	The subnet mask.
--enable	Enables the DHCP server.

Table 1: Terms and definitions

4.2 DHCP server configuration check

Your DHCP server is created, you can look at the list of the different configurations with the command below:

```
.\VBoxManage.exe list dhcpservers
```

```
c:\Program Files\Oracle\VirtualBox>.\VBoxManage.exe list dhcpservers
```

Figure 5: Command execution

```
NetworkName:    Isolated_Network
Dhcpd IP:       10.0.0.2
LowerIPAddress: 10.0.0.10
UpperIPAddress: 10.0.0.20
NetworkMask:    255.255.255.0
Enabled:        Yes
Global Configuration:
  minLeaseTime:    default
  defaultLeaseTime: default
  maxLeaseTime:    default
  Forced options:  None
  Suppressed opts.: None
  1/legacy: 255.255.255.0
Groups:          None
Individual Configs: None
c:\Program Files\Oracle\VirtualBox>
```

Figure 6: Result of executed command

4.3 Check IP address assignment via DHCP server

We're going to check that our machines have received an IP address from the DHCP server we've configured. As you can see in the picture below, our attacking virtual machine has received an IP address.

```
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
   link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
   inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
   inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
   link/ether 08:00:27:58:1d:08 brd ff:ff:ff:ff:ff:ff
   inet 10.0.0.14/24 brd 10.0.0.255 scope global dynamic noprefixroute eth0
       valid_lft 533sec preferred_lft 533sec
   inet6 fe80::a00:27ff:fe58:1d08/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
```

Figure 7: Attacker's virtual machine IP address

To get the network configuration of our virtual machine, we used the following command:

```
ip a
```

Now we're going to scan our network, to get the IP address of the target's virtual machine. To do this, we'll use the following command:

```
nmap -sn 10.0.0.0/24
```

Here's the result:

```
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled.
Nmap scan report for 10.0.0.2
Host is up (0.00020s latency).
MAC Address: 08:00:27:91:88:4F (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Nmap scan report for 10.0.0.11
Host is up (0.0030s latency).
MAC Address: 08:00:27:9B:81:5F (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Nmap scan report for 10.0.0.14
Host is up.
Nmap done: 256 IP addresses (3 hosts up) scanned in 1.97 seconds
```

Figure 8: Local network scan

To scan the network, we used the **NMAP** tool. It's a tool that allows you to scan the network for things like: discover active hosts and services, identify vulnerabilities and much more. In our case, we used the **-sn** option, to discover hosts on the network.

As you can see, we've found the IP addresses of our attacker's virtual machine (**10.0.0.14**) and the target's virtual machine (**10.0.0.11**). We can check that the target's virtual machine is communicating with the attacker's virtual machine using the following command:

```
ping -c3 10.0.0.11
```

```
PING 10.0.0.11 (10.0.0.11) 56(84) bytes of data.  
64 bytes from 10.0.0.11: icmp_seq=1 ttl=255 time=0.461 ms  
64 bytes from 10.0.0.11: icmp_seq=2 ttl=255 time=0.400 ms  
64 bytes from 10.0.0.11: icmp_seq=3 ttl=255 time=0.553 ms  
  
--- 10.0.0.11 ping statistics ---  
3 packets transmitted, 3 received, 0% packet loss, time 2049ms  
rtt min/avg/max/mdev = 0.400/0.471/0.553/0.062 ms
```

Figure 9: Ping the target virtual machine

As you can see, the target virtual machine responds to the ping. We'll also check that our virtual machines are not communicating with the external network (Internet).

```
ping: connect: Network is unreachable
```

Figure 10: Ping to external network (Internet)

The configuration of our hermetic network of virtual machines is now complete, and you can carry out your various tests without impacting your local network.

5 THANKS

Keep learning and you'll reach your goals. Thank you for reading this content.

See you soon.

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